

VULNERABILITY ASSESSMENT REPORT

Prepared For

Sun TV

Prepared By

IARM Information Security

Report Date

2 June 2026

2 June 2026

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IARM



Table of Contents

1.0 Document Control ... 2.0 About Us ... 3.0 Engagement Summary ... 3.1 Project Overview ... 3.2 Goals ... 3.3 Approach ... 3.4 Framework ... 3.5 Reassessment Guidelines ... 3.5.1 Out of Scope ... 3.5.2 Post Assessment Clean-up ... 3.5.3 Severity Rating Scale ... 4.0 Executive Summary of Findings ... 4.1 Vulnerability Assessment Risk Score ... 4.2 Risk Pie Chart by Vulnerabilities ... 5.0 Assessment Report ... 6.0 Conclusion ... 7.0 Appendix ... 7.1 Checklist Overview ...



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1.0 Document Control

Client Confidentiality

This document contains **Client Confidential information** and may not

Document Version Control

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2.0 About Us



IARM



CREST

IARM is a distinguished cyber security organization that provides an ex

IARM holds **CREST accreditation** for penetration testing services — a

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3.0 Engagement Summary

3.1 Project Overview

The organization **Sun TV** enlisted **IARM Information Security** for a V

Scope Centre

Project ID	IARM-2026-2544
Assessment Type	Gray Box
Time Frame	20 May 2026 – 1 June 2026
Report Date	2 June 2026
Scoped IPs Count	30

3.2 Goals

- **Identify and assess vulnerabilities:** Perform automated scans on i
- **Validate and Prioritize findings:** Analyze scan results to confirm m
- **Provide recommendations for mitigation:** Suggest appropriate re
- **Ensure compliance and resilience:** Support the organization in st

3.3 Approach

IARM Information Security Pvt. Ltd. conducted a **Vulnerability Assessment**

The assessment was performed using automated scanning techniques to

The scanning activities were carried out in a controlled manner within

The assessment approach broadly included:

- Identification of in-scope systems and devices
- Execution of automated vulnerability scans across selected assets
- Detection of known vulnerabilities and security gaps
- Consolidation of findings for reporting and risk visibility
- Recommendation of remediation measures to address identified issues

This approach provided a baseline understanding of the security risks and

3.4 Framework

IARM Information Security Pvt. Ltd.'s Vulnerability Assessment methodology

The methodology is based on the following industry standards:



OWASP

Open Web Application Security Project

3.5 Reassessment Guidelines

A one-time retest is optional. The retest shall be performed on all identified vulnerabilities. The customer must confirm the completion of remediation for all identified vulnerabilities. Any retest not conducted within **60 Days** from the date of submission of the final report shall be considered a new assessment.

3.5.1 Out of Scope

- Assessment is limited to what is mentioned in the scope of work as provided by the customer.
- Fixing any vulnerability during the Vulnerability assessment is not part of the scope.
- Any other services or solutions not mentioned in the project scope are not part of the assessment.
- External Web Application Penetration Testing is not part of the Assessment.
- Any post incident and log review Assessment is not part of the Assessment.

3.5.2 Post Assessment Clean-up

Upon completion of the Vulnerability Assessment, any access provided to the customer's systems shall be removed.

3.5.3 Severity Rating Scale

The assessment team used defined criteria to evaluate and rate the findings.

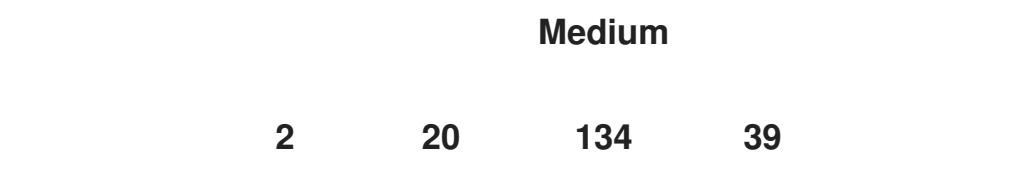
CRITICAL	System-level compromise can result from this vulnerability, posing a critical risk to the system, including urgent patching and mitigation to prevent potential full system takeover.
HIGH	Exploitation of this vulnerability poses a considerable risk of enabling system disruption. It is strongly recommended to plan and apply the necessary patches in a timely manner.
MEDIUM	The identified vulnerabilities pose a moderate risk or require more conditions such as user interaction for exploitation. A remediation plan should be developed and implemented in a prompt manner.
LOW	The identified vulnerabilities are not directly exploitable; however, they may be leveraged in combination with other weaknesses. It is recommended to define a remediation plan and schedule maintenance windows for updates.
INFO	The identified findings do not represent a direct risk, but they provide information that could be useful for future analysis. They may affect the confidentiality, integrity, or availability of the system, but the risk is low or non-existent when combined with other factors.

4.0 Executive Summary of Findings

4.1 Vulnerability Assessment Risk Score

For each identified finding, IARM Information Security applies a structured risk scoring model. The objective of this scoring model is to provide a clear and consistent

4.2 Risk Pie Chart by Vulnerabilities



C1	SSL Version 2 and 3 Protocol Detection
C2	SSL Version 2 and 3 Protocol Detection
H1	SSL Medium Strength Cipher Suites Supported

H2	SSL Medium Strength Cipher Suites Supported
H3	SSL Medium Strength Cipher Suites Supported
H4	SSL Medium Strength Cipher Suites Supported
H5	SSL Medium Strength Cipher Suites Supported
H6	SSL Medium Strength Cipher Suites Supported
H7	SSL Medium Strength Cipher Suites Supported
H8	rsh Service Detection
H9	SNMP Agent Default Community Name (
H10	SSL Medium Strength Cipher Suites Supported
H11	SSL Medium Strength Cipher Suites Supported
H12	SSL Medium Strength Cipher Suites Supported
H13	SSL Medium Strength Cipher Suites Supported
H14	SSL Medium Strength Cipher Suites Supported
H15	SSL Medium Strength Cipher Suites Supported

H16	SNMP Agent Default Community Name (
H17	rsh Service Detection
H18	SNMP Agent Default Community Name (
H19	SSL Medium Strength Cipher Suites Supported
H20	SNMP Agent Default Community Name (
M1	Network Time Protocol (NTP) Mode 6 S
M2	SSL Certificate with Wrong Hostnan
M3	SSL Certificate Cannot Be Trusted
M4	SSL Self-Signed Certificate
M5	SMB Signing not required
M6	SSL Certificate with Wrong Hostnan
M7	SSL Certificate Cannot Be Trusted
M8	SSL Self-Signed Certificate
M9	SMB Signing not required

M10	SSL Certificate Cannot Be Trusted
M11	SSL Self-Signed Certificate
M12	SMB Signing not required
M13	SMB Signing not required
M14	SMB Signing not required
M15	SSL Certificate Cannot Be Trusted
M16	SSL Self-Signed Certificate
M17	SMB Signing not required
M18	TLS Version 1.0 Protocol Detection
M19	TLS Version 1.1 Depreciated Protocol
M20	SMB Signing not required
M21	SSL Certificate Cannot Be Trusted
M22	SSL Self-Signed Certificate
M23	SMB Signing not required

M24	Terminal Services Doesn't Use Network Level Authentication
M25	TLS Version 1.0 Protocol Detection
M26	TLS Version 1.1 Deprecated Protocol
M27	SMB Signing not required
M28	SSL Certificate Cannot Be Trusted
M29	SSL Self-Signed Certificate
M30	SMB Signing not required
M31	TLS Version 1.0 Protocol Detection
M32	TLS Version 1.1 Deprecated Protocol
M33	SSL Certificate Cannot Be Trusted
M34	SSL Self-Signed Certificate
M35	SMB Signing not required
M36	TLS Version 1.0 Protocol Detection
M37	SSL Certificate Cannot Be Trusted

M38	SSL Self-Signed Certificate
M39	SMB Signing not required
M40	TLS Version 1.0 Protocol Detection
M41	TLS Version 1.1 Deprecated Protocol
M42	SSL Certificate Cannot Be Trusted
M43	SSL Self-Signed Certificate
M44	SMB Signing not required
M45	TLS Version 1.0 Protocol Detection
M46	TLS Version 1.1 Deprecated Protocol
M47	SSL Certificate Cannot Be Trusted
M48	SSL Self-Signed Certificate
M49	SMB Signing not required
M50	SMB Signing not required
M51	SSL Certificate Cannot Be Trusted

M52	SSL Self-Signed Certificate
M53	SMB Signing not required
M54	Terminal Services Doesn't Use Network Level Authentication
M55	TLS Version 1.0 Protocol Detection
M56	TLS Version 1.1 Depreciated Protocol
M57	SMB Signing not required
M58	SSL Certificate Cannot Be Trusted
M59	SSL Self-Signed Certificate
M60	SMB Signing not required
M61	Terminal Services Doesn't Use Network Level Authentication
M62	TLS Version 1.0 Protocol Detection
M63	TLS Version 1.1 Depreciated Protocol
M64	SSL Certificate Cannot Be Trusted
M65	SSL Self-Signed Certificate

M66	SMB Signing not required
M67	TLS Version 1.0 Protocol Detection
M68	TLS Version 1.1 Deprecated Protocol
M69	mDNS Detection (Remote Network)
M70	SSL Certificate Signed Using Weak Hashing
M71	Unencrypted Telnet Server
M72	SSL Certificate Cannot Be Trusted
M73	SSL Self-Signed Certificate
M74	SSL RC4 Cipher Suites Supported (Bar M)
M75	SNMP 'GETBULK' Reflection DDoS
M76	TLS Version 1.0 Protocol Detection
M77	TLS Version 1.1 Deprecated Protocol
M78	Network Time Protocol (NTP) Mode 6 S
M79	SMB Signing not required

M80	SSL Certificate Cannot Be Trusted
M81	SSL Self-Signed Certificate
M82	SMB Signing not required
M83	TLS Version 1.0 Protocol Detection
M84	TLS Version 1.1 Deprecated Protocol
M85	SSL Certificate Cannot Be Trusted
M86	SSL Self-Signed Certificate
M87	SMB Signing not required
M88	TLS Version 1.0 Protocol Detection
M89	TLS Version 1.1 Deprecated Protocol
M90	SSL Certificate Cannot Be Trusted
M91	SSL Self-Signed Certificate
M92	SMB Signing not required
M93	TLS Version 1.0 Protocol Detection

M94	TLS Version 1.1 Deprecated Protocol
M95	SSL Certificate Cannot Be Trusted
M96	SSL Self-Signed Certificate
M97	TLS Version 1.0 Protocol Detection
M98	TLS Version 1.1 Deprecated Protocol
M99	SSL Certificate Cannot Be Trusted
M100	SSL Self-Signed Certificate
M101	SMB Signing not required
M102	TLS Version 1.0 Protocol Detection
M103	TLS Version 1.1 Deprecated Protocol
M104	SSL Certificate Cannot Be Trusted
M105	SSL Self-Signed Certificate
M106	SMB Signing not required
M107	TLS Version 1.0 Protocol Detection

M108	TLS Version 1.1 Deprecated Protocol
M109	SSL Certificate Cannot Be Trusted
M110	SSL Self-Signed Certificate
M111	SMB Signing not required
M112	TLS Version 1.0 Protocol Detection
M113	TLS Version 1.1 Deprecated Protocol
M114	SSL Certificate Cannot Be Trusted
M115	SSL Self-Signed Certificate
M116	SMB Signing not required
M117	TLS Version 1.0 Protocol Detection
M118	TLS Version 1.1 Deprecated Protocol
M119	SSL Certificate Cannot Be Trusted
M120	SSL Self-Signed Certificate
M121	Terminal Services Doesn't Use Network Level Authent

M122	TLS Version 1.0 Protocol Detection
M123	TLS Version 1.1 Deprecated Protocol
M124	SNMP 'GETBULK' Reflection DDoS
M125	mDNS Detection (Remote Network)
M126	SSL Certificate Signed Using Weak Hashing
M127	SSL Certificate Cannot Be Trusted
M128	SSL Self-Signed Certificate
M129	SSL RC4 Cipher Suites Supported (Bar M)
M130	SNMP 'GETBULK' Reflection DDoS
M131	TLS Version 1.0 Protocol Detection
M132	TLS Version 1.1 Deprecated Protocol
M133	SNMP 'GETBULK' Reflection DDoS
M134	Network Time Protocol (NTP) Mode 6 S
L1	TLS Version 1.1 Deprecated Protocol

L2	ICMP Timestamp Request Remote Date Di
L3	ICMP Timestamp Request Remote Date Di
L4	ICMP Timestamp Request Remote Date Di
L5	ICMP Timestamp Request Remote Date Di
L6	ICMP Timestamp Request Remote Date Di
L7	ICMP Timestamp Request Remote Date Di
L8	ICMP Timestamp Request Remote Date Di
L9	ICMP Timestamp Request Remote Date Di
L10	ICMP Timestamp Request Remote Date Di
L11	ICMP Timestamp Request Remote Date Di
L12	ICMP Timestamp Request Remote Date Di
L13	ICMP Timestamp Request Remote Date Di
L14	ICMP Timestamp Request Remote Date Di
L15	ICMP Timestamp Request Remote Date Di

L16	ICMP Timestamp Request Remote Date Di
L17	ICMP Timestamp Request Remote Date Di
L18	ICMP Timestamp Request Remote Date Di
L19	ICMP Timestamp Request Remote Date Di
L20	ICMP Timestamp Request Remote Date Di
L21	ICMP Timestamp Request Remote Date Di
L22	ICMP Timestamp Request Remote Date Di
L23	SSL Certificate Chain Contains RSA Keys Less T
L24	SSLv3 Padding Oracle On Downgraded Legacy Encryption
L25	ICMP Timestamp Request Remote Date Di
L26	ICMP Timestamp Request Remote Date Di
L27	ICMP Timestamp Request Remote Date Di
L28	ICMP Timestamp Request Remote Date Di
L29	ICMP Timestamp Request Remote Date Di

L30	ICMP Timestamp Request Remote Date Di
L31	ICMP Timestamp Request Remote Date Di
L32	ICMP Timestamp Request Remote Date Di
L33	ICMP Timestamp Request Remote Date Di
L34	ICMP Timestamp Request Remote Date Di
L35	ICMP Timestamp Request Remote Date Di
L36	ICMP Timestamp Request Remote Date Di
L37	SSL Certificate Chain Contains RSA Keys Less T
L38	SSLv3 Padding Oracle On Downgraded Legacy Encryption
L39	ICMP Timestamp Request Remote Date Di

5.0 Assessment Report

The vulnerabilities below were identified and verified by IARM Informa

5.1 SSL Version 2 and 3 Protocol Detection

Critical	10.7.2.240	54443/tcp	C1

Description:

The remote service accepts connections encrypted using SSL 2.0 and/or

- An insecure padding scheme with CBC ciphers.
- Insecure session renegotiation and resumption schemes.

An attacker can exploit these flaws to conduct man-in-the-middle attac

Although SSL/TLS has a secure means for choosing the highest support

NIST has determined that SSL 3.0 is no longer acceptable for secure co

Proof of Concept

- SSLv3 is enabled and the server supports at least one

Explanation: TLS 1.0 and SSL 3.0 cipher suites may be u

Medium Strength Ciphers (> 64-bit and < 112-bit key, c

Name	Code	KEX
-----	-----	---
EDH-RSA-DES-CBC3-SHA		DHE
DES-CBC3-SHA		RSA

High Strength Ciphers (>= 112-bit key)

Name	Code	KEX
-----	-----	---
DHE-RSA-AES128-SHA		DHE
DHE-RSA-AES128-SHA256		DHE
DHE-RSA-AES256-SHA		DHE
DHE-RSA-AES256-SHA256		DHE
ECDHE-RSA-AES128-SHA256		ECDHE
ECDHE-RSA-AES256-SHA384		ECDHE
AES128-SHA		RSA
RSA-AES128-SHA256		RSA
AES256-SHA		RSA
RSA-AES256-SHA256		RSA
RC4-SHA		RSA

The fields above are :

{Tenable ciphername}
{Cipher ID code}

```
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
MAC={message authentication code}
{export flag}
```

Recommendation

Consult the application's documentation to disable SSL 2.0 and 3.0. Use

References

- <https://www.schneier.com/academic/paperfiles/paper-ssl.pdf>
- <http://www.nessus.org/u?b06c7e95>
- <http://www.nessus.org/u?247c4540>
- <https://www.openssl.org/~bodo/ssl-poodle.pdf>
- <http://www.nessus.org/u?5d15ba70>
- <https://www.imperialviolet.org/2014/10/14/poodle.html>
- <https://tools.ietf.org/html/rfc7507>
- <https://tools.ietf.org/html/rfc7568>

5.2 SSL Version 2 and 3 Protocol Detection

Critical	10.7.5.241	54443/tcp	C2

Description:

The remote service accepts connections encrypted using SSL 2.0 and/or

- An insecure padding scheme with CBC ciphers.
- Insecure session renegotiation and resumption schemes.

An attacker can exploit these flaws to conduct man-in-the-middle attack

Although SSL/TLS has a secure means for choosing the highest support

NIST has determined that SSL 3.0 is no longer acceptable for secure com

Proof of Concept

- SSLv3 is enabled and the server supports at least one

Explanation: TLS 1.0 and SSL 3.0 cipher suites may be v

Medium Strength Ciphers (> 64-bit and < 112-bit key, c

Name	Code	KEX
-----	-----	---
EDH-RSA-DES-CBC3-SHA		DHE
DES-CBC3-SHA		RSA

High Strength Ciphers (>= 112-bit key)

Name	Code	KEX
-----	-----	---
DHE-RSA-AES128-SHA		DHE
DHE-RSA-AES128-SHA256		DHE
DHE-RSA-AES256-SHA		DHE
DHE-RSA-AES256-SHA256		DHE
ECDHE-RSA-AES128-SHA256		ECDHE
ECDHE-RSA-AES256-SHA384		ECDHE
AES128-SHA		RSA
RSA-AES128-SHA256		RSA
AES256-SHA		RSA
RSA-AES256-SHA256		RSA
RC4-SHA		RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
MAC={message authentication code}
```

```
{export flag}
```

Recommendation

Consult the application's documentation to disable SSL 2.0 and 3.0. Use

References

- <https://www.schneier.com/academic/paperfiles/paper-ssl.pdf>
- <http://www.nessus.org/u?b06c7e95>
- <http://www.nessus.org/u?247c4540>
- <https://www.openssl.org/~bodo/ssl-poodle.pdf>
- <http://www.nessus.org/u?5d15ba70>
- <https://www.imperialviolet.org/2014/10/14/poodle.html>
- <https://tools.ietf.org/html/rfc7507>
- <https://tools.ietf.org/html/rfc7568>

5.3 SSL Medium Strength Cipher Suites Support

High	10.7.2.57	3389/tcp	H1

Description:

The remote host supports the use of SSL ciphers that offer medium strength encryption.

Note that it is considerably easier to circumvent medium strength encryption than high strength encryption.

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.4 SSL Medium Strength Cipher Suites Support

High	10.7.2.71	3389/tcp	H2

Description:

The remote host supports the use of SSL ciphers that offer medium strength encryption.

Note that it is considerably easier to circumvent medium strength encryption than high strength encryption.

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.5 SSL Medium Strength Cipher Suites Support

High	10.7.2.86	3389/tcp	H3

Description:

The remote host supports the use of SSL ciphers that offer medium strength encryption.

Note that it is considerably easier to circumvent medium strength encryption than high strength encryption.

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.6 SSL Medium Strength Cipher Suites Support

High	10.7.2.92	3389/tcp	H4

Description:

The remote host supports the use of SSL ciphers that offer medium strength encryption.

Note that it is considerably easier to circumvent medium strength encryption than strong encryption.

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.7 SSL Medium Strength Cipher Suites Support

High	10.7.2.174	3389/tcp	H5

Description:

The remote host supports the use of SSL ciphers that offer medium strength encryption.

Note that it is considerably easier to circumvent medium strength encryption than strong encryption.

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.8 SSL Medium Strength Cipher Suites Support

High	10.7.2.190	3389/tcp	H6

Description:

The remote host supports the use of SSL ciphers that offer medium strength encryption.

Note that it is considerably easier to circumvent medium strength encryption than high strength encryption.

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.9 SSL Medium Strength Cipher Suites Support

High	10.7.2.200	3389/tcp	H7

Description:

The remote host supports the use of SSL ciphers that offer medium strength encryption.

Note that it is considerably easier to circumvent medium strength encryption than high strength encryption.

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.10 rsh Service Detection

High	10.7.2.240	514/tcp	H8

Description:

The rsh service is running on the remote host. This service is vulnerable to the .rhosts or rhosts.equiv files.

Recommendation

Comment out the 'rsh' line in /etc/inetd.conf and restart the inetd process.

References

- [CVE-1999-0651](#)

5.11 SNMP Agent Default Community Name

High	10.7.2.240	161/udp	H9

Description:

It is possible to obtain the default community name of the remote SNMP agent.

An attacker may use this information to gain more knowledge about the target network.

Proof of Concept

```
The remote SNMP server replies to the following default
string :

public
```

Recommendation

Disable the SNMP service on the remote host if you do not use it. Either by using the command:

References

- [CVE-1999-0517](#)

5.12 SSL Medium Strength Cipher Suites Sup

High	10.7.2.240	54443/tcp	H10

Description:

The remote host supports the use of SSL ciphers that offer medium stre

Note that it is considerably easier to circumvent medium strength encry

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
EDH-RSA-DES-CBC3-SHA	0x00, 0x16	DHE
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
```

```
Encrypt={symmetric encryption method}  
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.13 SSL Medium Strength Cipher Suites Sup

High	10.7.5.58	3389/tcp	H11

Description:

The remote host supports the use of SSL ciphers that offer medium stre

Note that it is considerably easier to circumvent medium strength encry

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.14 SSL Medium Strength Cipher Suites Sup

High	10.7.5.78	3389/tcp	H12

Description:

The remote host supports the use of SSL ciphers that offer medium stre

Note that it is considerably easier to circumvent medium strength encry

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.15 SSL Medium Strength Cipher Suites Sup

High	10.7.5.83	3389/tcp	H13

Description:

The remote host supports the use of SSL ciphers that offer medium stre

Note that it is considerably easier to circumvent medium strength encry

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.16 SSL Medium Strength Cipher Suites Sup

High	10.7.5.109	3389/tcp	H14

Description:

The remote host supports the use of SSL ciphers that offer medium stre

Note that it is considerably easier to circumvent medium strength encry

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.17 SSL Medium Strength Cipher Suites Sup

High	10.7.5.116	3389/tcp	H15

Description:

The remote host supports the use of SSL ciphers that offer medium stre

Note that it is considerably easier to circumvent medium strength encry

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
Encrypt={symmetric encryption method}
```

```
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.18 SNMP Agent Default Community Name

High	10.7.5.240	161/udp	H16

Description:

It is possible to obtain the default community name of the remote SNMP
An attacker may use this information to gain more knowledge about the

Proof of Concept

```
The remote SNMP server replies to the following default
string :

public
```

Recommendation

Disable the SNMP service on the remote host if you do not use it. Either

References

- [CVE-1999-0517](#)

5.19 rsh Service Detection

High	10.7.5.241	514/tcp	H17

Description:

The rsh service is running on the remote host. This service is vulnerable to the .rhosts or rhosts.equiv files.

Recommendation

Comment out the 'rsh' line in /etc/inetd.conf and restart the inetd process.

References

- [CVE-1999-0651](#)

5.20 SNMP Agent Default Community Name

High	10.7.5.241	161/udp	H18

Description:

It is possible to obtain the default community name of the remote SNMP agent.

An attacker may use this information to gain more knowledge about the target network.

Proof of Concept

```
The remote SNMP server replies to the following default
string :

public
```

Recommendation

Disable the SNMP service on the remote host if you do not use it. Either by using the command:

References

- [CVE-1999-0517](#)

5.21 SSL Medium Strength Cipher Suites Sup

High	10.7.5.241	54443/tcp	H19

Description:

The remote host supports the use of SSL ciphers that offer medium stre

Note that it is considerably easier to circumvent medium strength encry

Proof of Concept

Medium Strength Ciphers (> 64-bit and < 112-bit key, or

Name	Code	KEX
-----	-----	---
EDH-RSA-DES-CBC3-SHA	0x00, 0x16	DHE
DES-CBC3-SHA	0x00, 0x0A	RSA

The fields above are :

```
{Tenable ciphername}
{Cipher ID code}
Kex={key exchange}
Auth={authentication}
```

```
Encrypt={symmetric encryption method}  
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application if possible to avoid use of medium

References

- <http://www.nessus.org/u?df5555f5>
- <https://sweet32.info>

5.22 SNMP Agent Default Community Name

High	10.7.5.242	161/udp	H20

Description:

It is possible to obtain the default community name of the remote SNMP
An attacker may use this information to gain more knowledge about the

Proof of Concept

```
The remote SNMP server replies to the following default
string :

public
```

Recommendation

Disable the SNMP service on the remote host if you do not use it. Either

References

- [CVE-1999-0517](#)

5.23 Network Time Protocol (NTP) Mode 6 S

Medium	10.7.2.1	123/udp	M1

Description:

The remote NTP server responds to mode 6 queries. Devices that respon

Proof of Concept

```
Nessus elicited the following response from the remote
host by sending an NTP mode 6 query :

'system="cisco", leap=0, stratum=5, rootdelay=205.23,
rootdispersion=164.02, peer=20512, refid=10.100.0.5,
reftime=0xEDBA7EE3.ADD295E7, poll=8, clock=0xEDBA7FB0.5,
phase=0.012, freq=-170.10, error=0.14'
```

Recommendation

Restrict NTP mode 6 queries.

References

- <https://www.nessus.org/u?18e62de1>

- <https://www.virtuesecurity.com/kb/ntp-mode-6-vulnerabilities/>

5.24 SSL Certificate with Wrong Hostname

Medium	10.7.2.12	7070/tcp	M2

Description:

The 'commonName' (CN) attribute of the SSL certificate presented for t

Proof of Concept

The identities known by Nessus are :

10.7.2.12

10.7.2.12

The Common Name in the certificate is :

AnyDesk Client

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.25 SSL Certificate Cannot Be Trusted

Medium	10.7.2.12	7070/tcp	M3

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=AnyDesk Client  
| -Issuer  : CN=AnyDesk Client
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.26 SSL Self-Signed Certificate

Medium	10.7.2.12	7070/tcp	M4

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=AnyDesk Client

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.27 SMB Signing not required

Medium	10.7.2.12	445/tcp	M5

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.28 SSL Certificate with Wrong Hostname

Medium	10.7.2.13	7070/tcp	M6

Description:

The 'commonName' (CN) attribute of the SSL certificate presented for t

Proof of Concept

The identities known by Nessus are :

mrc-sun-sops044.sunnetwork.in
10.7.2.13

The Common Name in the certificate is :

AnyDesk Client

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.29 SSL Certificate Cannot Be Trusted

Medium	10.7.2.13	3389/tcp	M7

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS044.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS044.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.30 SSL Self-Signed Certificate

Medium	10.7.2.13	3389/tcp	M8

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS044.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.31 SMB Signing not required

Medium	10.7.2.13	445/tcp	M9

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.32 SSL Certificate Cannot Be Trusted

Medium	10.7.2.15	7070/tcp	M10

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=AnyDesk Client  
| -Issuer  : CN=AnyDesk Client
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.33 SSL Self-Signed Certificate

Medium	10.7.2.15	7070/tcp	M11

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=AnyDesk Client

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.34 SMB Signing not required

Medium	10.7.2.15	445/tcp	M12

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.35 SMB Signing not required

Medium	10.7.2.18	445/tcp	M13

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.36 SMB Signing not required

Medium	10.7.2.52	445/tcp	M14

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.37 SSL Certificate Cannot Be Trusted

Medium	10.7.2.57	3389/tcp	M15

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-REC013.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-REC013.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.38 SSL Self-Signed Certificate

Medium	10.7.2.57	3389/tcp	M16

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-REC013.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.39 SMB Signing not required

Medium	10.7.2.57	445/tcp	M17

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.40 TLS Version 1.0 Protocol Detection

Medium	10.7.2.57	3389/tcp	M18

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated as of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.41 TLS Version 1.1 Deprecated Protocol

Medium	10.7.2.57	3389/tcp	M19

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to this attack.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to this attack.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.42 SMB Signing not required

Medium	10.7.2.59	445/tcp	M20

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.43 SSL Certificate Cannot Be Trusted

Medium	10.7.2.71	3389/tcp	M21

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS015.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS015.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.44 SSL Self-Signed Certificate

Medium	10.7.2.71	3389/tcp	M22

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS015.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.45 SMB Signing not required

Medium	10.7.2.71	445/tcp	M23

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.46 Terminal Services Doesn't Use Network

Medium	10.7.2.71	3389/tcp	M24

Description:

The remote Terminal Services is not configured to use Network Level A

Proof of Concept

Nessus was able to negotiate non-NLA (Network Level Autl

Recommendation

Enable Network Level Authentication (NLA) on the remote RDP server.

References

- <https://docs.microsoft.com/en-us/previous-versions/windows/it-pr>
- <http://www.nessus.org/u?e2628096>

5.47 TLS Version 1.0 Protocol Detection

Medium	10.7.2.71	3389/tcp	M25

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and should be disabled. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.48 TLS Version 1.1 Deprecated Protocol

Medium	10.7.2.71	3389/tcp	M26

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is not supported by TLS 1.2.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.49 SMB Signing not required

Medium	10.7.2.74	445/tcp	M27

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.50 SSL Certificate Cannot Be Trusted

Medium	10.7.2.84	3389/tcp	M28

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS035.sunnetwork.in
| -Issuer  : CN=MRC-SUN-SOPS035.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.51 SSL Self-Signed Certificate

Medium	10.7.2.84	3389/tcp	M29

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS035.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.52 SMB Signing not required

Medium	10.7.2.84	445/tcp	M30

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.53 TLS Version 1.0 Protocol Detection

Medium	10.7.2.84	3389/tcp	M31

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern browsers and operating systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.54 TLS Version 1.1 Deprecated Protocol

Medium	10.7.2.84	3389/tcp	M32

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.55 SSL Certificate Cannot Be Trusted

Medium	10.7.2.86	3389/tcp	M33

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS050.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS050.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.56 SSL Self-Signed Certificate

Medium	10.7.2.86	3389/tcp	M34

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS050.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.57 SMB Signing not required

Medium	10.7.2.86	445/tcp	M35

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.58 TLS Version 1.0 Protocol Detection

Medium	10.7.2.86	3389/tcp	M36

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern browsers and operating systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.59 SSL Certificate Cannot Be Trusted

Medium	10.7.2.92	3389/tcp	M37

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS039.sunnetwork.in  
| -Issuer : CN=MRC-SUN-SOPS039.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.60 SSL Self-Signed Certificate

Medium	10.7.2.92	3389/tcp	M38

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS039.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.61 SMB Signing not required

Medium	10.7.2.92	445/tcp	M39

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.62 TLS Version 1.0 Protocol Detection

Medium	10.7.2.92	3389/tcp	M40

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern browsers and operating systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.63 TLS Version 1.1 Deprecated Protocol

Medium	10.7.2.92	3389/tcp	M41

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

```
TLSh1.1 is enabled and the server supports at least one
```

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.64 SSL Certificate Cannot Be Trusted

Medium	10.7.2.116	3389/tcp	M42

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-PROG027.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-PROG027.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.65 SSL Self-Signed Certificate

Medium	10.7.2.116	3389/tcp	M43

Description:

The X.509 certificate chain for this service is not signed by a recognized authority. Note that this plugin does not check for certificate chains that end in a self-signed certificate.

Proof of Concept

The following certificate was found at the top of the certificate chain sent by the remote host, but is self-signed and was not found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-PROG027.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.66 SMB Signing not required

Medium	10.7.2.116	445/tcp	M44

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.67 TLS Version 1.0 Protocol Detection

Medium	10.7.2.116	3389/tcp	M45

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated as of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.68 TLS Version 1.1 Deprecated Protocol

Medium	10.7.2.116	3389/tcp	M46

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.69 SSL Certificate Cannot Be Trusted

Medium	10.7.2.136	3389/tcp	M47

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-FV09.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-FV09.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.70 SSL Self-Signed Certificate

Medium	10.7.2.136	3389/tcp	M48

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-FV09.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.71 SMB Signing not required

Medium	10.7.2.136	445/tcp	M49

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.72 SMB Signing not required

Medium	10.7.2.138	445/tcp	M50

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.73 SSL Certificate Cannot Be Trusted

Medium	10.7.2.174	3389/tcp	M51

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS045.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS045.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.74 SSL Self-Signed Certificate

Medium	10.7.2.174	3389/tcp	M52

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and w
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS045.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.75 SMB Signing not required

Medium	10.7.2.174	445/tcp	M53

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.76 Terminal Services Doesn't Use Network

Medium	10.7.2.174	3389/tcp	M54

Description:

The remote Terminal Services is not configured to use Network Level A

Proof of Concept

Nessus was able to negotiate non-NLA (Network Level Autl

Recommendation

Enable Network Level Authentication (NLA) on the remote RDP server.

References

- <https://docs.microsoft.com/en-us/previous-versions/windows/it-pr>
- <http://www.nessus.org/u?e2628096>

5.77 TLS Version 1.0 Protocol Detection

Medium	10.7.2.174	3389/tcp	M55

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

```
TLsv1 is enabled and the server supports at least one c
```

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.78 TLS Version 1.1 Deprecated Protocol

Medium	10.7.2.174	3389/tcp	M56

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

```
TLSv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.
```

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.79 SMB Signing not required

Medium	10.7.2.179	445/tcp	M57

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.80 SSL Certificate Cannot Be Trusted

Medium	10.7.2.190	3389/tcp	M58

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS066.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS066.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.81 SSL Self-Signed Certificate

Medium	10.7.2.190	3389/tcp	M59

Description:

The X.509 certificate chain for this service is not signed by a recognized authority. Note that this plugin does not check for certificate chains that end in a self-signed certificate.

Proof of Concept

The following certificate was found at the top of the certificate chain sent by the remote host, but is self-signed and was not found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS066.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.82 SMB Signing not required

Medium	10.7.2.190	445/tcp	M60

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.83 Terminal Services Doesn't Use Network

Medium	10.7.2.190	3389/tcp	M61

Description:

The remote Terminal Services is not configured to use Network Level A

Proof of Concept

Nessus was able to negotiate non-NLA (Network Level Autl

Recommendation

Enable Network Level Authentication (NLA) on the remote RDP server.

References

- <https://docs.microsoft.com/en-us/previous-versions/windows/it-pr>
- <http://www.nessus.org/u?e2628096>

5.84 TLS Version 1.0 Protocol Detection

Medium	10.7.2.190	3389/tcp	M62

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated as of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

```
TLsv1 is enabled and the server supports at least one c
```

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.85 TLS Version 1.1 Deprecated Protocol

Medium	10.7.2.190	3389/tcp	M63

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.86 SSL Certificate Cannot Be Trusted

Medium	10.7.2.200	3389/tcp	M64

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-FV015.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-FV015.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.87 SSL Self-Signed Certificate

Medium	10.7.2.200	3389/tcp	M65

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-FV015.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.88 SMB Signing not required

Medium	10.7.2.200	445/tcp	M66

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.89 TLS Version 1.0 Protocol Detection

Medium	10.7.2.200	3389/tcp	M67

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern browsers and operating systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

```
TLsv1 is enabled and the server supports at least one c
```

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.90 TLS Version 1.1 Deprecated Protocol

Medium	10.7.2.200	3389/tcp	M68

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is not supported by TLS 1.2.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.91 mDNS Detection (Remote Network)

Medium	10.7.2.240	5353/udp	M69

Description:

The remote service understands the Bonjour (also known as ZeroConf or mDNS)

This plugin attempts to discover mDNS used by hosts that are not on the local network

Proof of Concept

Nessus was able to extract the following information :

- mDNS hostname : Billing.local.
- Advertised services :
 - o Service name : Billing._printer._tcp.local.
Port number : 515
 - o Service name : Billing._pdl-datastream._tcp.local.
Port number : 9100
 - o Service name : Billing._ipp._tcp.local.
Port number : 631
 - o Service name : Billing._ipps._tcp.local.
Port number : 443

- o Service name : Billing._uscan._tcp.local.
Port number : 80
- o Service name : Billing._uscans._tcp.local.
Port number : 443
- o Service name : Billing._http._tcp.local.
Port number : 80

Recommendation

Filter incoming traffic to UDP port 5353, if desired.

5.92 SSL Certificate Signed Using Weak Hash

Medium	10.7.2.240	54443/tcp	M70

Description:

The remote service uses an SSL certificate chain that has been signed u

Note that this plugin reports all SSL certificate chains signed with SHA-

Note that certificates in the chain that are contained in the Nessus CA c

Proof of Concept

The following certificates were part of the certificate
the remote host, but contain hashes that are considered

```
Subject          : C=JP/ST=Kanagawa/L=Ebina/O=RICOH O
Signature Algorithm : MD5 With RSA Encryption
Valid From       : Sep 06 15:00:00 2016 GMT
Valid To         : Sep 01 15:00:00 2036 GMT
Raw PEM certificate :
-----BEGIN CERTIFICATE-----
MIICQDCCAamgAwIBAAIJAJGMzHWdc5RxMA0GCSqGSIb3DQEBAUAMGI
+YIHu4naqK6hw/iP04NW53uWBBkmVhicgKzYblsqLLJYKmLJ/T2je9Z
Page 157 of 285
```

-----END CERTIFICATE-----

Recommendation

Contact the Certificate Authority to have the SSL certificate reissued.

References

- <https://tools.ietf.org/html/rfc3279>
- <http://www.nessus.org/u?9bb87bf2>
- <http://www.nessus.org/u?e120eea1>
- <http://www.nessus.org/u?5d894816>
- <http://www.nessus.org/u?51db68aa>
- <http://www.nessus.org/u?9dc7bfba>

5.93 Unencrypted Telnet Server

Medium	10.7.2.240	23/tcp	M71

Description:

The remote host is running a Telnet server over an unencrypted channel.

Using Telnet over an unencrypted channel is not recommended as login credentials are sent in plain text.

SSH is preferred over Telnet since it protects credentials from eavesdropping.

Proof of Concept

```
Nessus collected the following banner from the remote Telnet server:

----- snip -----

RICOH Maintenance Shell.

User access verification.

login:

----- snip -----
```

Recommendation

Disable the Telnet service and use SSH instead.

5.94 SSL Certificate Cannot Be Trusted

Medium	10.7.2.240	443/tcp	M72

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : C=gb/CN=Billing  
| -Issuer  : C=gb/CN=Billing
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.95 SSL Self-Signed Certificate

Medium	10.7.2.240	443/tcp	M73

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : C=gb/CN=Billing

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.96 SSL RC4 Cipher Suites Supported (Bar M

Medium	10.7.2.240	54443/tcp	M74

Description:

The remote host supports the use of RC4 in one or more cipher suites. 7

If plaintext is repeatedly encrypted (e.g., HTTP cookies), and an attacker

Proof of Concept

List of RC4 cipher suites supported by the remote server:

High Strength Ciphers (>= 112-bit key)

Name	Code	KEX
-----	-----	---
RC4-SHA	0x00, 0x05	RSA

The fields above are :

{Tenable ciphername}

{Cipher ID code}

Kex={key exchange}

```
Auth={authentication}  
Encrypt={symmetric encryption method}  
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application, if possible, to avoid use of RC4 cipher.

References

- <https://www.rc4nomore.com/>
- <http://www.nessus.org/u?ac7327a0>
- <http://cr.yp.to/talks/2013.03.12/slides.pdf>
- <http://www.isg.rhul.ac.uk/tls/>
- https://www.imperva.com/docs/HII_Attacking_SSL_when_using_RC4

5.97 SNMP 'GETBULK' Reflection DDoS

Medium	10.7.2.240	161/udp	M75

Description:

The remote SNMP daemon is responding with a large amount of data to

Proof of Concept

Nessus was able to determine the SNMP service can be abused for a
Reflection DDoS attack :

Request size (bytes) : 42
Response size (bytes) : 1398

Recommendation

Disable the SNMP service on the remote host if you do not use it. Other

References

- <http://www.nessus.org/u?8b551b5c>

5.98 TLS Version 1.0 Protocol Detection

Medium	10.7.2.240	54443/tcp	M76

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern browsers and operating systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.99 TLS Version 1.1 Deprecated Protocol

Medium	10.7.2.240	443/tcp	M77

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLSh1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.100 Network Time Protocol (NTP) Mode 6

Medium	10.7.5.1	123/udp	M78

Description:

The remote NTP server responds to mode 6 queries. Devices that respond

Proof of Concept

Nessus elicited the following response from the remote host by sending an NTP mode 6 query :

```
'system="cisco", leap=0, stratum=5, rootdelay=205.14, rootdispersion=142.82, peer=39744, refid=10.100.0.5, reftime=0xEDBA8175.EB852140, poll=6, clock=0xEDBA818D.7, phase=9.487, freq=204.64, error=0.37'
```

Recommendation

Restrict NTP mode 6 queries.

References

- <https://www.nessus.org/u?18e62de1>

- <https://www.virtuesecurity.com/kb/ntp-mode-6-vulnerabilities/>

5.101 SMB Signing not required

Medium	10.7.5.24	445/tcp	M79

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.102 SSL Certificate Cannot Be Trusted

Medium	10.7.5.58	3389/tcp	M80

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS065.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS065.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.103 SSL Self-Signed Certificate

Medium	10.7.5.58	3389/tcp	M81

Description:

The X.509 certificate chain for this service is not signed by a recognized authority. Note that this plugin does not check for certificate chains that end in a self-signed certificate.

Proof of Concept

The following certificate was found at the top of the certificate chain sent by the remote host, but is self-signed and was not found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS065.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.104 SMB Signing not required

Medium	10.7.5.58	445/tcp	M82

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.105 TLS Version 1.0 Protocol Detection

Medium	10.7.5.58	3389/tcp	M83

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern browsers and operating systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.106 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.58	3389/tcp	M84

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.107 SSL Certificate Cannot Be Trusted

Medium	10.7.5.78	3389/tcp	M85

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS37.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS37.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.108 SSL Self-Signed Certificate

Medium	10.7.5.78	3389/tcp	M86

Description:

The X.509 certificate chain for this service is not signed by a recognized authority. Note that this plugin does not check for certificate chains that end in a self-signed certificate.

Proof of Concept

The following certificate was found at the top of the certificate chain sent by the remote host, but is self-signed and was not found in the list of known certificate authorities :

```
| -Subject : CN=MRC-SUN-SOPS37.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.109 SMB Signing not required

Medium	10.7.5.78	445/tcp	M87

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.110 TLS Version 1.0 Protocol Detection

Medium	10.7.5.78	3389/tcp	M88

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern browsers and operating systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.111 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.78	3389/tcp	M89

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.112 SSL Certificate Cannot Be Trusted

Medium	10.7.5.83	3389/tcp	M90

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS056.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS056.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.113 SSL Self-Signed Certificate

Medium	10.7.5.83	3389/tcp	M91

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS056.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.114 SMB Signing not required

Medium	10.7.5.83	445/tcp	M92

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.115 TLS Version 1.0 Protocol Detection

Medium	10.7.5.83	3389/tcp	M93

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is no longer supported by many systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.116 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.83	3389/tcp	M94

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is not supported by TLS 1.2.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.117 SSL Certificate Cannot Be Trusted

Medium	10.7.5.94	3389/tcp	M95

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS030.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS030.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.118 SSL Self-Signed Certificate

Medium	10.7.5.94	3389/tcp	M96

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS030.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.119 TLS Version 1.0 Protocol Detection

Medium	10.7.5.94	3389/tcp	M97

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern browsers and operating systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

`TLSv1 is enabled and the server supports at least one c`

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.120 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.94	3389/tcp	M98

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.121 SSL Certificate Cannot Be Trusted

Medium	10.7.5.109	3389/tcp	M99

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-PROG032.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-PROG032.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.122 SSL Self-Signed Certificate

Medium	10.7.5.109	3389/tcp	M100

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-PROG032.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.123 SMB Signing not required

Medium	10.7.5.109	445/tcp	M101

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.124 TLS Version 1.0 Protocol Detection

Medium	10.7.5.109	3389/tcp	M102

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is no longer supported by many systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

```
TLsv1 is enabled and the server supports at least one c
```

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.125 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.109	3389/tcp	M103

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

```
TLShv1.1 is enabled and the server supports at least one cipher suite that is not supported by TLS 1.2
```

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.126 SSL Certificate Cannot Be Trusted

Medium	10.7.5.116	3389/tcp	M104

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS038.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS038.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.127 SSL Self-Signed Certificate

Medium	10.7.5.116	3389/tcp	M105

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS038.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.128 SMB Signing not required

Medium	10.7.5.116	445/tcp	M106

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.129 TLS Version 1.0 Protocol Detection

Medium	10.7.5.116	3389/tcp	M107

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is no longer supported by many systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

```
TLsv1 is enabled and the server supports at least one c
```

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.130 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.116	3389/tcp	M108

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.131 SSL Certificate Cannot Be Trusted

Medium	10.7.5.118	3389/tcp	M109

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-PROG043.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-PROG043.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.132 SSL Self-Signed Certificate

Medium	10.7.5.118	3389/tcp	M110

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-PROG043.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.133 SMB Signing not required

Medium	10.7.5.118	445/tcp	M111

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.134 TLS Version 1.0 Protocol Detection

Medium	10.7.5.118	3389/tcp	M112

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is not supported by modern browsers and operating systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

```
TLsv1 is enabled and the server supports at least one c
```

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.135 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.118	3389/tcp	M113

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is not supported by TLS 1.2.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.136 SSL Certificate Cannot Be Trusted

Medium	10.7.5.121	3389/tcp	M114

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SRY-PROG05.sunnetwork.in  
| -Issuer  : CN=MRC-SRY-PROG05.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.137 SSL Self-Signed Certificate

Medium	10.7.5.121	3389/tcp	M115

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : CN=MRC-SRY-PROG05.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.138 SMB Signing not required

Medium	10.7.5.121	445/tcp	M116

Description:

Signing is not required on the remote SMB server. An unauthenticated,

Recommendation

Enforce message signing in the host's configuration. On Windows, this i

References

- <http://www.nessus.org/u?df39b8b3>
- <http://technet.microsoft.com/en-us/library/cc731957.aspx>
- <http://www.nessus.org/u?74b80723>
- <https://www.samba.org/samba/docs/current/man-html/smb.conf.5>
- <http://www.nessus.org/u?a3cac4ea>

5.139 TLS Version 1.0 Protocol Detection

Medium	10.7.5.121	3389/tcp	M117

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and is no longer supported by many modern systems. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

```
TLsv1 is enabled and the server supports at least one c
```

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.140 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.121	3389/tcp	M118

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is not supported by TLS 1.2.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.141 SSL Certificate Cannot Be Trusted

Medium	10.7.5.126	3389/tcp	M119

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : CN=MRC-SUN-SOPS033.sunnetwork.in  
| -Issuer  : CN=MRC-SUN-SOPS033.sunnetwork.in
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.142 SSL Self-Signed Certificate

Medium	10.7.5.126	3389/tcp	M120

Description:

The X.509 certificate chain for this service is not signed by a recognized authority. Note that this plugin does not check for certificate chains that end in a self-signed certificate.

Proof of Concept

The following certificate was found at the top of the certificate chain sent by the remote host, but is self-signed and was not found in the list of known certificate authorities :

| -Subject : CN=MRC-SUN-SOPS033.sunnetwork.in

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.143 Terminal Services Doesn't Use Network

Medium	10.7.5.126	3389/tcp	M121

Description:

The remote Terminal Services is not configured to use Network Level A

Proof of Concept

```
Nessus was able to negotiate non-NLA (Network Level Autl
```

Recommendation

Enable Network Level Authentication (NLA) on the remote RDP server.

References

- <https://docs.microsoft.com/en-us/previous-versions/windows/it-pr>
- <http://www.nessus.org/u?e2628096>

5.144 TLS Version 1.0 Protocol Detection

Medium	10.7.5.126	3389/tcp	M122

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and should be disabled. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

```
TLSSv1 is enabled and the server supports at least one c
```

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.145 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.126	3389/tcp	M123

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.146 SNMP 'GETBULK' Reflection DDoS

Medium	10.7.5.240	161/udp	M124

Description:

The remote SNMP daemon is responding with a large amount of data to

Proof of Concept

Nessus was able to determine the SNMP service can be abused for a
Reflection DDoS attack :

Request size (bytes) : 42
Response size (bytes) : 1818

Recommendation

Disable the SNMP service on the remote host if you do not use it. Other

References

- <http://www.nessus.org/u?8b551b5c>

5.147 mDNS Detection (Remote Network)

Medium	10.7.5.241	5353/udp	M125

Description:

The remote service understands the Bonjour (also known as ZeroConf o

This plugin attempts to discover mDNS used by hosts that are not on th

Proof of Concept

Nessus was able to extract the following information :

- mDNS hostname : Sales.local.
- Advertised services :
 - o Service name : RICOH IM 2702 [58387973A933].
Port number : 515
 - o Service name : RICOH IM 2702 [58387973A933].
Port number : 9100
 - o Service name : RICOH IM 2702 [58387973A933].
Port number : 631
 - o Service name : RICOH IM 2702 [58387973A933].
Port number : 443

- o Service name : RICOH IM 2702 [58387973A933] ..
Port number : 80
- o Service name : RICOH IM 2702 [58387973A933] ..
Port number : 443
- o Service name : RICOH IM 2702 [58387973A933] ..
Port number : 80

Recommendation

Filter incoming traffic to UDP port 5353, if desired.

5.148 SSL Certificate Signed Using Weak Hash

Medium	10.7.5.241	54443/tcp	M126

Description:

The remote service uses an SSL certificate chain that has been signed u

Note that this plugin reports all SSL certificate chains signed with SHA-

Note that certificates in the chain that are contained in the Nessus CA c

Proof of Concept

The following certificates were part of the certificate
the remote host, but contain hashes that are considered

```
Subject          : C=JP/ST=Kanagawa/L=Ebina/O=RICOH C
Signature Algorithm : MD5 With RSA Encryption
Valid From       : Sep 06 15:00:00 2016 GMT
Valid To         : Sep 01 15:00:00 2036 GMT
Raw PEM certificate :
-----BEGIN CERTIFICATE-----
MIICQDCCAamgAwIBAAIJAJGMzHWdc5RxMA0GCSqGSIb3DQEBAUAMGI
+YIHu4naqK6hw/iP04NW53uWBBkmVhicgKzYblsqLLJYKmLJ/T2je9Z
Page 228 of 285
```

-----END CERTIFICATE-----

Recommendation

Contact the Certificate Authority to have the SSL certificate reissued.

References

- <https://tools.ietf.org/html/rfc3279>
- <http://www.nessus.org/u?9bb87bf2>
- <http://www.nessus.org/u?e120eea1>
- <http://www.nessus.org/u?5d894816>
- <http://www.nessus.org/u?51db68aa>
- <http://www.nessus.org/u?9dc7bfba>

5.149 SSL Certificate Cannot Be Trusted

Medium	10.7.5.241	443/tcp	M127

Description:

The server's X.509 certificate cannot be trusted. This situation can occur

- First, the top of the certificate chain sent by the server might not be
- Second, the certificate chain may contain a certificate that is not va
- Third, the certificate chain may contain a signature that either didn

If the remote host is a public host in production, any break in the chain

Proof of Concept

The following certificate was at the top of the certifi
chain sent by the remote host, but it is signed by an un
certificate authority :

```
| -Subject : C=gb/CN=Sales  
| -Issuer  : C=gb/CN=Sales
```

Recommendation

Purchase or generate a proper SSL certificate for this service.

References

- <https://www.itu.int/rec/T-REC-X.509/en>
- <https://en.wikipedia.org/wiki/X.509>

5.150 SSL Self-Signed Certificate

Medium	10.7.5.241	443/tcp	M128

Description:

The X.509 certificate chain for this service is not signed by a recognized

Note that this plugin does not check for certificate chains that end in a

Proof of Concept

The following certificate was found at the top of the c
chain sent by the remote host, but is self-signed and wa
found in the list of known certificate authorities :

| -Subject : C=gb/CN=Sales

Recommendation

Purchase or generate a proper SSL certificate for this service.

5.151 SSL RC4 Cipher Suites Supported (Bar

Medium	10.7.5.241	54443/tcp	M129

Description:

The remote host supports the use of RC4 in one or more cipher suites. 7

If plaintext is repeatedly encrypted (e.g., HTTP cookies), and an attacker

Proof of Concept

List of RC4 cipher suites supported by the remote server:

High Strength Ciphers (>= 112-bit key)

Name	Code	KEX
-----	-----	---
RC4-SHA	0x00, 0x05	RSA

The fields above are :

{Tenable ciphername}

{Cipher ID code}

Kex={key exchange}

```
Auth={authentication}  
Encrypt={symmetric encryption method}  
MAC={message authentication code}  
{export flag}
```

Recommendation

Reconfigure the affected application, if possible, to avoid use of RC4 cipher.

References

- <https://www.rc4nomore.com/>
- <http://www.nessus.org/u?ac7327a0>
- <http://cr.yp.to/talks/2013.03.12/slides.pdf>
- <http://www.isg.rhul.ac.uk/tls/>
- https://www.imperva.com/docs/HII_Attacking_SSL_when_using_RC4

5.152 SNMP 'GETBULK' Reflection DDoS

Medium	10.7.5.241	161/udp	M130

Description:

The remote SNMP daemon is responding with a large amount of data to

Proof of Concept

Nessus was able to determine the SNMP service can be abused for a
Reflection DDoS attack :

Request size (bytes) : 42
Response size (bytes) : 1392

Recommendation

Disable the SNMP service on the remote host if you do not use it. Other

References

- <http://www.nessus.org/u?8b551b5c>

5.153 TLS Version 1.0 Protocol Detection

Medium	10.7.5.241	54443/tcp	M131

Description:

The remote service accepts connections encrypted using TLS 1.0. TLS 1.0 is deprecated and should be disabled. As of March 31, 2020, Endpoints that aren't enabled for TLS 1.2 and higher are non-compliant with PCI DSS v3.2. PCI DSS v3.2 requires that TLS 1.0 be disabled entirely by June 30, 2019.

Proof of Concept

```
TLsv1 is enabled and the server supports at least one c
```

Recommendation

Enable support for TLS 1.2 and 1.3, and disable support for TLS 1.0.

References

- <https://tools.ietf.org/html/draft-ietf-tls-oldversions-deprecate-00>

5.154 TLS Version 1.1 Deprecated Protocol

Medium	10.7.5.241	443/tcp	M132

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLShv1.1 is enabled and the server supports at least one cipher suite that is vulnerable to downgrade attacks.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.155 SNMP 'GETBULK' Reflection DDoS

Medium	10.7.5.242	161/udp	M133

Description:

The remote SNMP daemon is responding with a large amount of data to

Proof of Concept

Nessus was able to determine the SNMP service can be abused for a
Reflection DDoS attack :

Request size (bytes) : 42
Response size (bytes) : 1815

Recommendation

Disable the SNMP service on the remote host if you do not use it. Other

References

- <http://www.nessus.org/u?8b551b5c>

5.156 Network Time Protocol (NTP) Mode 6

Medium	10.7.10.1	123/udp	M134

Description:

The remote NTP server responds to mode 6 queries. Devices that respon

Proof of Concept

```
Nessus elicited the following response from the remote
  host by sending an NTP mode 6 query :

'system="cisco", leap=0, stratum=5, rootdelay=204.96,
rootdispersion=144.44, peer=20512, refid=10.100.0.5,
reftime=0xEDBA8D23.B7F810F7, poll=6, clock=0xEDBA8D25.0
phase=-0.079, freq=-218.67, error=0.05'
```

Recommendation

Restrict NTP mode 6 queries.

References

- <https://www.nessus.org/u?18e62de1>

- <https://www.virtuesecurity.com/kb/ntp-mode-6-vulnerabilities/>

5.157 TLS Version 1.1 Deprecated Protocol

Low	10.7.2.86	3389/tcp	L1

Description:

The remote service accepts connections encrypted using TLS 1.1. TLS 1.1 is deprecated as of March 31, 2020, Endpoints that are not enabled for TLS 1.2 and higher are vulnerable to downgrade attacks.

Proof of Concept

TLSEsv1.1 is enabled and the server supports at least one cipher suite that is not supported by TLS 1.2.

Recommendation

Enable support for TLS 1.2 and/or 1.3, and disable support for TLS 1.1.

References

- <https://datatracker.ietf.org/doc/html/rfc8996>
- <http://www.nessus.org/u?c8ae820d>

5.158 ICMP Timestamp Request Remote Dat

Low	10.7.2.1	icmp	L2

Description:

The remote host answers to an ICMP timestamp request. This allows an

Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP tim

References

- CVE-1999-0524

5.159 ICMP Timestamp Request Remote Dat

Low	10.7.2.12	icmp	L3

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.160 ICMP Timestamp Request Remote Dat

Low	10.7.2.13	icmp	L4

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.161 ICMP Timestamp Request Remote Dat

Low	10.7.2.15	icmp	L5

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.162 ICMP Timestamp Request Remote Dat

Low	10.7.2.18	icmp	L6

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.163 ICMP Timestamp Request Remote Dat

Low	10.7.2.24	icmp	L7

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.164 ICMP Timestamp Request Remote Dat

Low	10.7.2.57	icmp	L8

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.165 ICMP Timestamp Request Remote Dat

Low	10.7.2.59	icmp	L9

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is 1

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.166 ICMP Timestamp Request Remote Dat

Low	10.7.2.71	icmp	L10

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.167 ICMP Timestamp Request Remote Dat

Low	10.7.2.74	icmp	L11

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.168 ICMP Timestamp Request Remote Dat

Low	10.7.2.84	icmp	L12

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.169 ICMP Timestamp Request Remote Dat

Low	10.7.2.86	icmp	L13

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.170 ICMP Timestamp Request Remote Dat

Low	10.7.2.92	icmp	L14

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.171 ICMP Timestamp Request Remote Dat

Low	10.7.2.116	icmp	L15

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.172 ICMP Timestamp Request Remote Dat

Low	10.7.2.136	icmp	L16

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.173 ICMP Timestamp Request Remote Dat

Low	10.7.2.138	icmp	L17

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.174 ICMP Timestamp Request Remote Dat

Low	10.7.2.174	icmp	L18

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.175 ICMP Timestamp Request Remote Dat

Low	10.7.2.179	icmp	L19

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.176 ICMP Timestamp Request Remote Dat

Low	10.7.2.190	icmp	L20

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.177 ICMP Timestamp Request Remote Dat

Low	10.7.2.200	icmp	L21

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.178 ICMP Timestamp Request Remote Dat

Low	10.7.2.240	icmp	L22

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The remote clock is synchronized with the local clock.

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.179 SSL Certificate Chain Contains RSA Ke

Low	10.7.2.240	54443/tcp	L23

Description:

At least one of the X.509 certificates sent by the remote host has a key

Some browser SSL implementations may reject keys less than 2048 bits

Note that Nessus will not flag root certificates with RSA keys less than

Proof of Concept

The following certificates were part of the certificate
sent by the remote host, but contain RSA keys that are
to be weak :

| -Subject : C=JP/ST=Kanagawa/L=Ebina/O=RICOH COM
| -RSA Key Length : 1024 bits

Recommendation

Replace the certificate in the chain with the RSA key less than 2048 bit

References

- https://www.cabforum.org/wp-content/uploads/Baseline_Requiren

5.180 SSLv3 Padding Oracle On Downgraded

Low	10.7.2.240	54443/tcp	L24

Description:

The remote host is affected by a man-in-the-middle (MitM) information

As long as a client and service both support SSLv3, a connection can be

The TLS Fallback SCSV mechanism prevents 'version rollback' attacks w

This is a vulnerability in the SSLv3 specification, not in any particular S

Proof of Concept

Nessus determined that the remote server supports SSLv3 cipher suite, indicating that this server is vulnerable

It appears that TLSv1 or newer is supported on the server. The TLS Fallback SCSV mechanism is not supported, allowing connection to SSLv3.

Recommendation

Disable SSLv3.

Services that must support SSLv3 should enable the TLS Fallback SCSV

References

- <https://www.imperialviolet.org/2014/10/14/poodle.html>
- <https://www.openssl.org/~bodo/ssl-poodle.pdf>
- <https://tools.ietf.org/html/draft-ietf-tls-downgrade-scsv-00>

5.181 ICMP Timestamp Request Remote Dat

Low	10.7.5.1	icmp	L25

Description:

The remote host answers to an ICMP timestamp request. This allows an

Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP timestamp

References

- CVE-1999-0524

5.182 ICMP Timestamp Request Remote Dat

Low	10.7.5.24	icmp	L26

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.183 ICMP Timestamp Request Remote Dat

Low	10.7.5.58	icmp	L27

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.184 ICMP Timestamp Request Remote Dat

Low	10.7.5.78	icmp	L28

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.185 ICMP Timestamp Request Remote Dat

Low	10.7.5.83	icmp	L29

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.186 ICMP Timestamp Request Remote Dat

Low	10.7.5.94	icmp	L30

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -7

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.187 ICMP Timestamp Request Remote Dat

Low	10.7.5.109	icmp	L31

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.188 ICMP Timestamp Request Remote Dat

Low	10.7.5.116	icmp	L32

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.189 ICMP Timestamp Request Remote Dat

Low	10.7.5.118	icmp	L33

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -2

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.190 ICMP Timestamp Request Remote Dat

Low	10.7.5.121	icmp	L34

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The ICMP timestamps seem to be in little endian format
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.191 ICMP Timestamp Request Remote Dat

Low	10.7.5.126	icmp	L35

Description:

The remote host answers to an ICMP timestamp request. This allows an
Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

This host returns non-standard timestamps (high bit is s
The ICMP timestamps might be in little endian format (n
The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP ti

References

- [CVE-1999-0524](#)

5.192 ICMP Timestamp Request Remote Dat

Low	10.7.5.241	icmp	L36

Description:

The remote host answers to an ICMP timestamp request. This allows an

Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The difference between the local and remote clocks is 64

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP timestamp

References

- CVE-1999-0524

5.193 SSL Certificate Chain Contains RSA Ke

Low	10.7.5.241	54443/tcp	L37

Description:

At least one of the X.509 certificates sent by the remote host has a key

Some browser SSL implementations may reject keys less than 2048 bits

Note that Nessus will not flag root certificates with RSA keys less than

Proof of Concept

The following certificates were part of the certificate
sent by the remote host, but contain RSA keys that are
to be weak :

| -Subject : C=JP/ST=Kanagawa/L=Ebina/O=RICOH COM
| -RSA Key Length : 1024 bits

Recommendation

Replace the certificate in the chain with the RSA key less than 2048 bit

References

- https://www.cabforum.org/wp-content/uploads/Baseline_Requiren

5.194 SSLv3 Padding Oracle On Downgraded

Low	10.7.5.241	54443/tcp	L38

Description:

The remote host is affected by a man-in-the-middle (MitM) information

As long as a client and service both support SSLv3, a connection can be

The TLS Fallback SCSV mechanism prevents 'version rollback' attacks w

This is a vulnerability in the SSLv3 specification, not in any particular S

Proof of Concept

Nessus determined that the remote server supports SSLv3 cipher suite, indicating that this server is vulnerable

It appears that TLSv1 or newer is supported on the server. The TLS Fallback SCSV mechanism is not supported, allowing connection to SSLv3.

Recommendation

Disable SSLv3.

Services that must support SSLv3 should enable the TLS Fallback SCSV

References

- <https://www.imperialviolet.org/2014/10/14/poodle.html>
- <https://www.openssl.org/~bodo/ssl-poodle.pdf>
- <https://tools.ietf.org/html/draft-ietf-tls-downgrade-scsv-00>

5.195 ICMP Timestamp Request Remote Dat

Low	10.7.10.1	icmp	L39

Description:

The remote host answers to an ICMP timestamp request. This allows an

Timestamps returned from machines running Windows Vista / 7 / 2008

Proof of Concept

The difference between the local and remote clocks is -3

Recommendation

Filter out the ICMP timestamp requests (13), and the outgoing ICMP tim

References

- CVE-1999-0524

6.0 Conclusion

Within the given time frame the Assessment exercise was conducted at

Information Security is not just an on-time effort, but also an ongoing a

The effort to identify Information security threats and associated breach

7.0 Appendix

7.1 Checklist Overview

1	Host Identification	
2	Open Port (Tcp/Udp)	
3	Service Detection	
4	Malware Detection	
5	Unpatched Software	
6	Known Vulnerabilities	
7	Security Misconfiguration	

8	Operating System Authentication Weaknesses
9	Insecure Protocols
10	Network Share Weaknesses
11	Common Mistakes & Configuration Weaknesses
12	Information Leakage
13	Database Misconfiguration
14	SMB/SNMP/LDAP Misconfigurations
15	Active Directory/Iam Misconfigurations
16	User Enumeration
17	Windows Management Instrumentation (WMI) queries
18	Active Directory Service Interfaces (ADSI) queries
19	The Security Account Manager (SAM) registry